

附录 1: 直纹曲面和桌脚边缘线图像绘制程序

```
%canshu
w=2.5;
r=sqrt(100+1)*w;
l=60-w;
d=l/2;
h=53-3;
theta=asin(h/l);

%shengcheng canshu
v=-25:25;
u=25:60;
[u,v]=meshgrid(u,v);

%qumian canshu fangcheng
x=v;
y=sqrt(r^2-v.^2)+(d*cos(theta)+w-sqrt(r^2-v.^2)).
*(u-sqrt(r^2-v.^2))./sqrt(d^2+w^2+r^2-v.^2-2*(d*
cos(theta)+w)*sqrt(r^2-v.^2)+2*d*w*cos(theta));
z=d*sin(theta)*(u-sqrt(r^2-v.^2))./sqrt(d^2+w^2+
r^2-v.^2-2*(d*cos(theta)+w)*sqrt(r^2-v.^2)+2*d*
w*cos(theta));
z=-z;

%cemian tuxiang
figure(1);
mesh(x,y,z);

%zhuojiao bianyuanxian
u=60;
v=-25:25;
x=v;
y=sqrt(r^2-v.^2)+(d*cos(theta)+w-sqrt(r^2-v.^2)).
*(u-sqrt(r^2-v.^2))./sqrt(d^2+w^2+r^2-v.^2-2*(d*
cos(theta)+w)*sqrt(r^2-v.^2)+2*d*w*cos(theta));
z=d*sin(theta)*(u-sqrt(r^2-v.^2))./sqrt(d^2+w^2+
r^2-v.^2-2*(d*cos(theta)+w)*sqrt(r^2-v.^2)+2*d*
w*cos(theta));
z=-z;
figure(2);
plot3(x,y,z,'LineWidth',2);
grid on;
```

附录 2: 圆形折叠桌的动态变化过程图绘制程序

```
%canshu
w=2.5;
r=sqrt(100+1)*w;
l=60-w;
d=l/2;
h=53-3;
theta=asin(h/l);

%zhuobian dian zuobiao
xc=-10:10;
xc=xc*w;
yc=sqrt(r^2-xc.^2);
zc=zeros(1,21);
```

```
%gangjin dian zuobiao
xg=-10:10;
xg=xg*w;
yg=d*cos(theta)*ones(1,21)+w;
zg=d*sin(theta)*ones(1,21);

%zhuobian dao gangjin de juli:
for i=1:21

dis(i)=norm([xc(i),yc(i),zc(i)]-[xg(i),yg(i),zg(i)]);
end

%kaicang dao banbian de juli:
for i=1:21
    margin(i)=60-yc(i)-dis(i);
end

%muban dingdian zuobiao
for i=1:21
    k=(margin(i)+dis(i))/dis(i);
    xd(i)=xc(i)+k*(xg(i)-xc(i));
    yd(i)=yc(i)+k*(yg(i)-yc(i));
    zd(i)=zc(i)+k*(zg(i)-zc(i));
end

figure(1); hold on;
plot3(xc,yc,zc,'*', 'LineWidth',2);
plot3(xg,yg,zg,'r', 'LineWidth',2);
for i=1:21

line([xc(i),xg(i)],[yc(i),yg(i)],[zc(i),zg(i)], 'LineWid
th',2);

line([xd(i),xg(i)],[yd(i),yg(i)],[zd(i),zg(i)], 'LineWi
dth',2);
end

figure(1); hold on;
plot3(xc,-yc,zc,'*');
plot3(xg,-yg,zg,'r');
for i=1:21

line([xc(i),xg(i)],[-yc(i),-yg(i)],[zc(i),zg(i)], 'LineW
idth',1, 'Color',[.2 .2 .2]);

line([xd(i),xg(i)],[-yd(i),-yg(i)],[zd(i),zg(i)], 'LineW
idth',1, 'Color',[.2 .2 .2]);
end

plot3(xc,yc,zc);plot3(xc,-yc,zc);
line([xc(1),xc(1)],[yc(1),-yc(1)],[zc(1),zc(1)], 'Line
Width',2);
line([xc(21),xc(21)],[yc(21),-yc(21)],[zc(21),zc(21
)], 'LineWidth',2);
view(3)

[X,Y,Z]=sphere(30);
```

```
X=25*X;Y=25*Y;Z=zeros(31);
surf(X,Y,Z);
colormap(spring);
alpha(.5)
shading interp; axis equal; axis off;
```

附录 3：多目标规划求解程序(问题 2)

```
global w h a W r x lamda;
w=2.5;h=70-3;a=1;W=80;lamda=1.5;r=sqrt(40*40
+2.5*2.5);
x=[2.5:2.5:40]';
ts0=[pi/4,h/2];
lb=[0,0];
ub=[pi/2,h];
ts=fmincon(@objfun,ts0,[],[],[],[],lb,ub,@confun)
```

目标函数文件objfun.m

```
function f=objfun(ts)
f=-sin(ts(1));
%di er mubiao
%l=w+h/sin(ts(1));
%d=l-ts(2);
%q=32.5-abs(x);
%len=sqrt(d^2-+w^2+q^2-2*(d*cos(ts(1))+w)*q+
2*d*w*cos(ts(1)))+q-d-w
%f=sum(len);
```

约束函数 confun.m

```
function [c,ceq]=confun(ts)
%ts=[theta,s];
```

```
global w h a W r x lamda;
l=w+h/sin(ts(1));
d=l-ts(2);
q=sqrt(r^2-x.^2);
len=sqrt(d^2-+w^2+r^2-x.^2-2*(d*cos(ts(1))+w)*
q+2*d*w*cos(ts(1)))+q-d-w;
```

```
c=[a*W-2*(w+h*cot(ts(1)));
```

```
-(q+(d*cos(ts(1))+w-q).*(l-q)./sqrt(d^2-+w^2+r^2-
x.^2-2*(d*cos(ts(1))+w)*q+2*d*w*cos(ts(1))));
len-ts(2)+lamda
];
ceq=[];
```

附录 4：问题 2 折叠桌示意图程序

```
%canshu
```

```
global w h a W r x lamda n;
w=2.5;h=70-3;a=1;W=80;lamda=5;r=sqrt(40*40+
2.5*2.5);
```

```
%youhua qiujie
```

```
x=[2.5:2.5:40]';
ts0=[pi/4,h/2];
lb=[0,0];
```

```
ub=[pi/2,h];
ts=fmincon(@objfun,ts0,[],[],[],[],lb,ub,@confun)
```

```
theta=ts(1); %youhua jieguo
s=ts(2); %youhua jieguo
l=w+h/sin(theta);
d=l-s;
n=80/2.5+1;
```

```
%zhuobian dian zuobiao
```

```
xc=-40:2.5:40;
yc=sqrt(r^2-xc.^2);
zc=zeros(1,n);
```

```
%gangjin dian zuobiao
```

```
xg=-40:2.5:40;
yg=d*cos(theta)*ones(1,n)+w;
zg=d*sin(theta)*ones(1,n);
```

```
%zhuobian dao gangjin de juli:
```

```
for i=1:n
```

```
dis(i)=norm([xc(i),yc(i),zc(i)]-[xg(i),yg(i),zg(i)]);
end
```

```
%kaicang dao banbian de juli:
```

```
for i=1:n
```

```
margin(i)=l-yc(i)-dis(i);
```

```
end
```

```
%muban dingdian zuobiao
```

```
for i=1:n
```

```
k=(margin(i)+dis(i))/dis(i);
```

```
xd(i)=xc(i)+k*(xg(i)-xc(i));
```

```
yd(i)=yc(i)+k*(yg(i)-yc(i));
```

```
zd(i)=zc(i)+k*(zg(i)-zc(i));
```

```
end
```

```
figure(1); hold on;
```

```
plot3(xc,yc,zc,'*');
```

```
plot3(xg,yg,zg,'r');
```

```
for i=1:n
```

```
line([xc(i),xg(i)],[yc(i),yg(i)],[zc(i),zg(i)],'LineWid
th',2);
```

```
line([xd(i),xg(i)],[yd(i),yg(i)],[zd(i),zg(i)],'LineWi
dth',2);
```

```
end
```

```
figure(1); hold on;
```

```
plot3(xc,-yc,zc,'*');
```

```
plot3(xg,-yg,zg,'r');
```

```
for i=1:n
```

```
line([xc(i),xg(i)],[yc(i),yg(i)],[zc(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);
```

```
line([xd(i),xg(i)],[-yd(i),-yg(i)],[zd(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);
end
```

```
plot3(xc,yc,zc);plot3(xc,-yc,zc);
line([xc(1),xc(1)],[yc(1),-yc(1)],[zc(1),zc(1)],'Line
Width',2);
line([xc(n),xc(n)],[yc(n),-yc(n)],[zc(n),zc(n)],'Line
Width',2);
view(3)
```

```
[X,Y,Z]=sphere(30);
X=l*X/2;Y=l*Y/2;Z=zeros(31);
surf(X,Y,Z);
colormap(spring);
alpha(.5)
shading interp; axis equal; axis off;
```

附录 5: 多目标规划求解程序(问题 3)

```
global w h a W x lamda;
w=2.5;h=60-3;a=1;W=60;lamda=5;
x=[2.5:2.5:30]';
%8xing, gaiwei:
%w=2.5;h=70-3;a=1;W=90;lamda=5;
%x=[2.5:2.5:45]';
ts0=[pi/4,h/2];
lb=[0,0];
ub=[pi/2,h];
ts=fmincon(@objfun,ts0,[],[],[],[],lb,ub,@confun)
```

```
confun.m:
function [c,ceq]=confun(ts)
%ts=[theta,s];
```

```
global w h a W x lamda;
l=w+h/sin(ts(1));
d=l-ts(2);
q=32.5-abs(x); %8xing, gaiwei:
q=sqrt(30^2-(x-15)^2);
len=sqrt(d^2-w^2+q^2-2*(d*cos(ts(1))+w)*q+2*
d*w*cos(ts(1)))+q-d-w;
```

```
c=[a*W-2*(w+h*cot(ts(1)));
```

```
-(q+(d*cos(ts(1))+w-q).*(l-q)./sqrt(d^2-w^2+r^2-
x.^2-2*(d*cos(ts(1))+w)*q+2*d*w*cos(ts(1))));
len-ts(2)+lamda
];
ceq=[];
```

```
objfun.m
function f=objfun(ts)
f=-sin(ts(1));
%di er mubiao
%l=w+h/sin(ts(1));
```

```
%d=l-ts(2);
%q=32.5-abs(x);
%len=sqrt(d^2-w^2+q^2-2*(d*cos(ts(1))+w)*q+
2*d*w*cos(ts(1)))+q-d-w
%f=sum(len);
```

附录 6: 正方形折叠桌的动态变化过程绘图程序

```
clear;
%canshu
w=2.5;h=60-3;a=1;W=60; n=W/2.5+1;s=32;
t=pi/3;
l=w+h/sin(t);
R=sqrt(l^2+30^2);
```

```
%zhuobian dian zuobiao
xc=-30:2.5:30;
yc=32.5-abs(xc);
zc=zeros(1,n);
```

```
xd=xc;yg=xc;
ll=sqrt(R^2-xc.^2);
d=l-s;
```

```
for m=1:8
tt=linspace(0,8,8);
theta=tt(m)*t/8;
```

```
%gangjin dian zuobiao
yg=d*cos(theta)*ones(1,n)+w;
zg=d*sin(theta)*ones(1,n);
```

```
%zhuobian dao gangjin de juli:
for i=1:n
```

```
dis(i)=norm([xc(i),yc(i),zc(i)]-[xg(i),yg(i),zg(i)]);
end
```

```
%kaicang dao banbian de juli:
margin=ll-yc-dis;
```

```
%muban dingdian zuobiao
for i=1:n
k=(margin(i)+dis(i))/dis(i);
yd(i)=yc(i)+k*(yg(i)-yc(i));
zd(i)=zc(i)+k*(zg(i)-zc(i));
end
```

```
subplot(2,4,m)
figure(1); hold on;
plot3(xc,yc,zc,'*');
plot3(xg,yg,zg,'r');
for i=1:n
```

```
line([xc(i),xg(i)],[yc(i),yg(i)],[zc(i),zg(i)],'LineWid
th',2);
```

```
line([xd(i),xg(i)],[yd(i),yg(i)],[zd(i),zg(i)],'LineWi
```

```

    dth',2);
    end

    plot3(xc,-yc,zc,'*');
    plot3(xg,-yg,zg,'r');
    for i=1:n

line([xc(i),xg(i)],[-yc(i),-yg(i)],[zc(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);

line([xd(i),xg(i)],[-yd(i),-yg(i)],[zd(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);
    end

    plot3(xc,yc,zc);plot3(xc,-yc,zc);

line([xc(1),xc(1)],[yc(1),-yc(1)],[zc(1),zc(1)],'Line
Width',2);

line([xc(n),xc(n)],[yc(n),-yc(n)],[zc(n),zc(n)],'Line
Width',2);
    view(3)

    X=[0 0;-1 1;0 0];Y=[1 1;0 0;-1 -1];
    X=32.5*X;Y=32.5*Y;Z=zeros(3,2);
    surf(X,Y,Z);
    colormap(spring);
    alpha(.5)
    shading interp; axis equal; axis off;
end
figure(2);
plot3(xd,yd,zd);plot3(xd,-yd,zd);

附录 7： 8 方形折叠桌的动态变化过程绘图程序
clear;
%canshu
w=2.5;h=70-3;W=90; n=W/2.5-1;s=40;
t=pi*70/180;

%zhuobian dian zuobiao
xc=-42.5:2.5:42.5;
yc=sqrt(30^2-(abs(xc)-15).^2);
zc=zeros(1,n);

l=yc(1)+h/sin(t);
xd=xc;xg=xc;
d=l-s;

for m=1:8
    theta=m*t/8;

    %gangjin dian zuobiao
    yg=d*cos(theta)*ones(1,n)+w;
    zg=d*sin(theta)*ones(1,n);

    %zhuobian dao gangjin de juli:
    for i=1:n

```

```

dis(i)=norm([xc(i),yc(i),zc(i)]-[xg(i),yg(i),zg(i)]);
    end

    %kaicang dao banbian de juli:
    margin=l-yc-dis;

    %muban dingdian zuobiao
    for i=1:n
        k=(margin(i)+dis(i))/dis(i);
        yd(i)=yc(i)+k*(yg(i)-yc(i));
        zd(i)=zc(i)+k*(zg(i)-zc(i));
    end

    subplot(2,4,m)
    figure(1); hold on;
    plot3(xc,yc,zc,'*');
    plot3(xg,yg,zg,'r');
    for i=1:n

line([xc(i),xg(i)],[yc(i),yg(i)],[zc(i),zg(i)],'LineWid
th',2);

line([xd(i),xg(i)],[yd(i),yg(i)],[zd(i),zg(i)],'LineWi
dth',2);
    end

    plot3(xc,-yc,zc,'*');
    plot3(xg,-yg,zg,'r');
    for i=1:n

line([xc(i),xg(i)],[-yc(i),-yg(i)],[zc(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);

line([xd(i),xg(i)],[-yd(i),-yg(i)],[zd(i),zg(i)],'LineW
idth',1,'Color',[.2 .2 .2]);
    end

    plot3(xc,yc,zc);plot3(xc,-yc,zc);

line([xc(1),xc(1)],[yc(1),-yc(1)],[zc(1),zc(1)],'Line
Width',2);

line([xc(n),xc(n)],[yc(n),-yc(n)],[zc(n),zc(n)],'Line
Width',2);
    view(3)

    [X,Y,Z]=sphere(30);
    X=30*X;Y=30*Y;Z=zeros(31);
    X=X+15;    surf(X,Y,Z);
    X=X-30;    surf(X,Y,Z);
    colormap(spring);
    alpha(.5)
    shading interp; axis equal; axis off;
end
figure(2);
plot3(xd,yd,zd);plot3(xd,-yd,zd);

```